An Analog Front-end IP for 13.56MHz RFID Interrogators

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Abstract— An analog front-end circuit for 13.56MHz RFID interrogators compatible with ISO14443, ISO15693 and ISO18000-3 Mode 1 RFID interrogators was designed and fabricated by using 0.35μm double poly CMOS process. The fabricated chip was operated at 3.3 volt single supply. The results of this work can be provided as reusable IPs in a form of hard or firm IPs for designing single chip 13.56MHz RFID interrogators.

I. INTRODUCTION

There has been a great interest in Radio Frequency Identification(RFID). RFID's using 13.56MHz carrier frequency have been most widely used in many fields such as E-commerce, credit card, mobile phones, personal administration, PDAs, ID tags, transportation and fare collection, etc[1-5]. The international standards for 13.56MHz RFID's are ISO14443 type A/type B, ISO15693 and ISO18000-3. Table 1 shows Key parameters of signal interface between transponders and interrogator for 13.56MHz RFID's. The system communicates in half duplex mode. The interrogator uses a carrier frequency(fc) of ISM band, 13.56MHz, both for transferring energy and data.

TABLE I
Key parameters of 13.56MHz RFID's signal interfaces between interrogator(Reader) and transponders (tags)[1-2]

ISO standard	14443 Type A	14443 Type B	15693	18000-3 Mode 1
Carrier Freq.(fc)	13.56MHz ± 7KHz			
Modulation (R → Tags)	100% ASK	10% ASK	Interrogator selects 10% or 100% ASK	
Coding (R → Tags)	Modified miller	NRZ	Pulse position modulation	
Baud rate (Tag →R)	106K	106K	1.65K or 26.5K	
Subcarrier Freq. (Tag (R)	847KHz (fc/16)	847KHz (fc/16)	one carrier: 423.75K(fs1) two carriers: fs1, 484.28K(fs2)	
Load modulation (Tag (R)	Manchester coding	BPSK	1 subcarrier or 2 subcarrier Manchester coding	

The ISO 14443 is a standard operating in proximity, where the communicating distance is less than 10 cm. For type-A interface, the interrogator sends 100% ASK modulated data encoded by modified Miller method. For the type-B interface, 10% ASK modulated signal of NRZ coded bits shall be transmitted to transponder.

The ISO15693 is a standard for vicinity cards, operating range is up to one meter. ISO18000-3 is a new standard for item management. The physical layer of air interface for ISO18000-3 Mode 1 is compatible with ISO15693.

The inductively coupled loop antenna of Reader and transponder form a transformer. The transponder can send data by load modulation, which is switching a load resistor or capacitor in the transponder with a subcarrier frequency.

The sub-carrier frequency of ISO14443 is 847 KHz, and the bit rate of communication is 106Kbps. For ISO14443 type A, the transponder modulates data by Manchester coding. On the other hand, the subcarrier is BPSK modulated, for the ISO14443 type-B interface.

For ISO15693 and 18000-3 Mode 1, tags respond to reader in subcarrier modulated Manchester code using one subcarrier or two subcarrier mode.

There were many publications regarding design of single chip set solution for RFID [3-8], however, there has been no technical papers regarding a design issue of ISO 14443, ISO 15693 and 18000-3 Mode 1 compatible RFID interrogators. The goal of this work is providing a reusable IP of analog front end circuit of 13.56MHz RFID interrogator for designing platform based single chip RFID readers.

II. DESIGN OF ANALOG FRONT END CIRCUIT

Fig.1 shows the block diagram of the RFID reader system. RFID reader system consists of a digital controller and an analog front end part. The analog front end part is composed of transmitting section for ASK modulator and antenna driving circuit, and receiving section for demodulating sub-carrier modulated input data from RFID. DIN is input data which shall be transmitted to a transponder. DIN is modified Miller coded, NRZ coded or pulse position modulated data according to the operating mode of ISO standard. RX_OUT is a demodulated output of the receiving circuit. The Fig. 1 shows a typical application where the antenna is directly connected to TX pin of the driver. The external antenna circuit consists of an EMC low pass filter, and an antenna matching circuit. The EMC filter is a low pass filter composed by L0 and C0, where the cut off frequency of the EMC filter was designed to 14.5MHz, whish is slightly above the carrier frequency. The resonance frequency of LC tank composed by C1, C2 and inductance of the loop antenna was tuned to carrier frequency of 13.56MHz. Where, C1 is a stray capacitance of the loop antenna. The value of C1 is depending on antenna's geometrical properties and material of a print circuit board. The matching circuit of the antenna must have a Q factor typically larger than 40, to eliminate the high frequency harmonics and to adjust bandwidth of receiving signal [8].

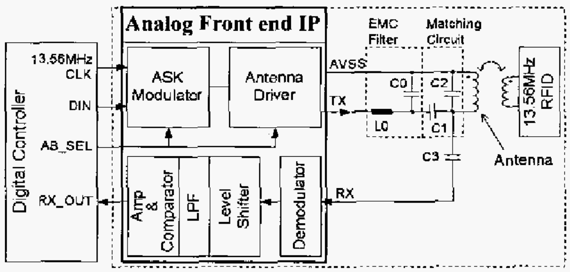

Fig.1 Block Diagram of a RFID Reader system.

A. ASK modulator and Antenna driver

Fig.2 shows the proposed circuit of the ASK modulator and Antenna Driver circuit of RFID Reader for 13.56MHz RFID interface. SEL is control signal selecting 100% ASK or 10% ASK mode. Output of MUX is CLK signal having 50% duty cycle by setting SEL to 'high', and NOR multiples CLK and DIN. Then, the data DIN is ASK modulated with CLK by the NOR gate, the inverter formed by MP and MN drives 100 % ASK modulated pulses to the TX node. The EMC filter and matching circuit are filtering high frequency harmonics of rectangular pulse of TX node.

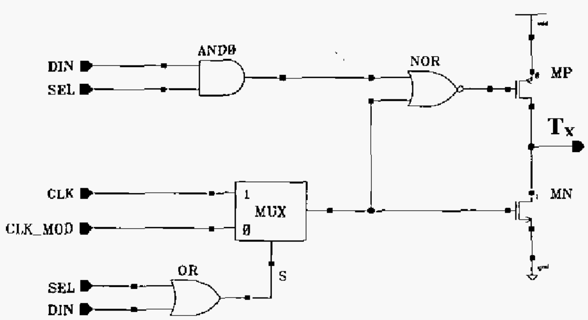

Fig.2 Proposed ASK modulator and Antenna Driver of RFID interrogator

If input of SEL is 'low', the output of MUX is whether CLK or CLK_MOD depending on transmitting data DIN. Where, CLK_MOD is duty cycle modulated clock signal. The Fourier coefficient of 13.56MHz frequency component is controlled by duty cycle of the clock applying to TX node. SPICE simulation results showed that antenna outputs are 10%, 20%, and 30% ASK modulated when duty cycle is 28 %, 20%, and 14%, respectively. Fig.3 shows proposed duty cycle modulation circuit. The CLK is delayed by the Delay Line circuit, where the delay time is controlled by S signal. So, the duty cycle of the CLK_MOD can be adjusted by the digital controller shown in Fig. 1. Fig. 4 shows simulated waveforms of DIN and output voltage on the antenna for 100% ASK mode, and for 10% mode

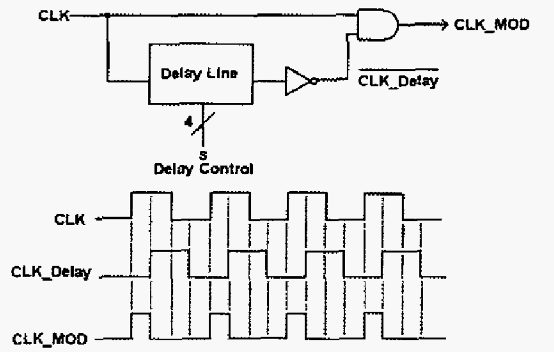

Fig.3 Duty cycle modulation circuit.

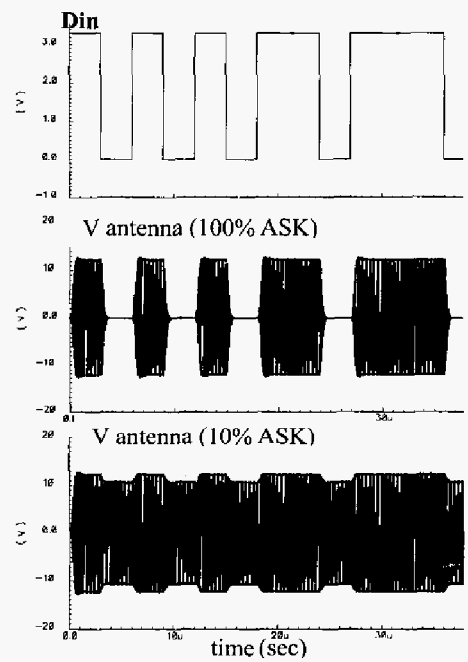

Fig.4 Simulated waveforms of D_{IN} and output voltage on the antenna for 100% ASK mode, and for 10% mode

B. Receiving circuit

Fig. 5 shows proposed circuit for the receiving block. The basic function of the receiving block is demodulating load modulated data in RX input, so that it reproduces a subcarrier modulated pulse signal, RX_OUT, and feeds it to the digital controller. The complete binary data can easily be produced in the digital controller. The voltage level of RX input is adjusted by voltage divider formed by RXL1 and R XL2, and an offset voltage is added by a clamper circuit so as to negative level is shifted to ground. The maximum amplitude of antennal signal is larger than 30 volt. The main purpose of the voltage divider is protecting mixer from a high voltage signal form the antenna. The designed ratio of the voltage divider is about

1/5. Negative input is not acceptable inside of the circuit, because we use only single positive supply as VDD. Moreover, we have been aiming a reusable IP for SoC solution of RFID interrogator so that a complicated technology such as triple well process is not suitable for analog IP design since process migration is important.

The mixer made of CMOS transmission gate multiplies CLK and incoming RF signal. The R_X input is a signal from the antenna, so the carrier of R_X signal is synchronized with CLK by themselves. The DC offset level of the mixer output is adjusted to half VDD in the level adjustment circuit before feeding to the filter and comparator circuitry. The low pass filters in the receiving section are suppressing high frequency noise where the cut off band is about 800 KHz. A high pass filter with cut off frequency of about 1.6MHz formed by C6, R10 and R11 is used to eliminate low frequency spurious signal.

The mid-band gain of low pass filter is about 10, and the gain of the inverting amplifier composed by R11 and R12 is about 10, so the overall gain of the receiving circuit was designed to about 100. A Schmitt trigger is used for the comparator. Fig.6 shows simulated waveforms of Manchester load modulated antenna voltage (R_X) and demodulated R_{X_OUT} voltage.

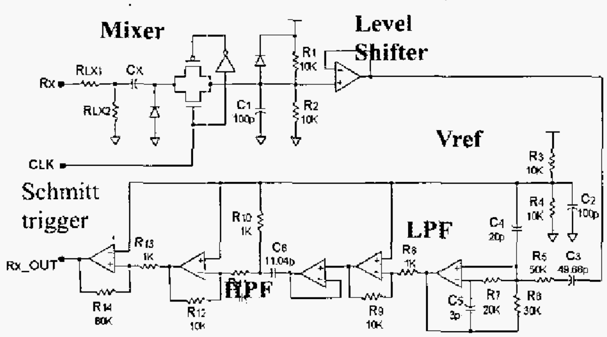

Fig. 5 Circuit diagram of the receiving block.

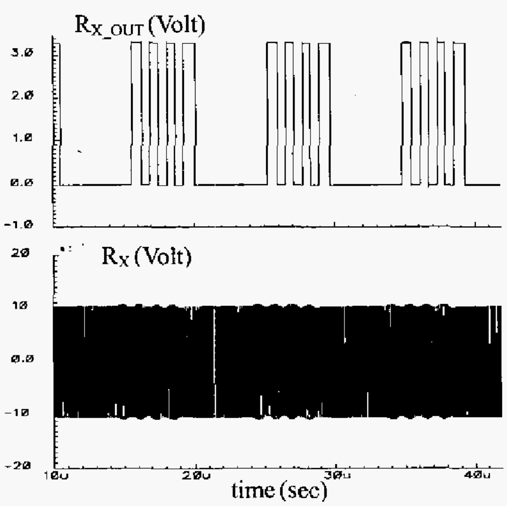

Fig.6 Simulated waveforms of Manchester load modulated antenna voltage (R_X) and demodulated R_{X_OUT} voltage

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

The chip was designed in full custom and fabricated by using a 0.35μm double poly CMOS process. Fig. 7 shows the layout of the fabricated single chip 13.56MHz RFID Interrogator.

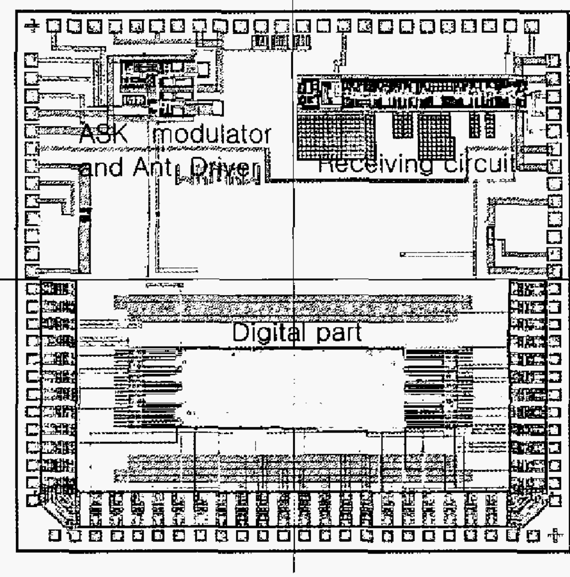

Fig. 7. Layout of the fabricated single chip 13.56MHz RFID Interrogator.

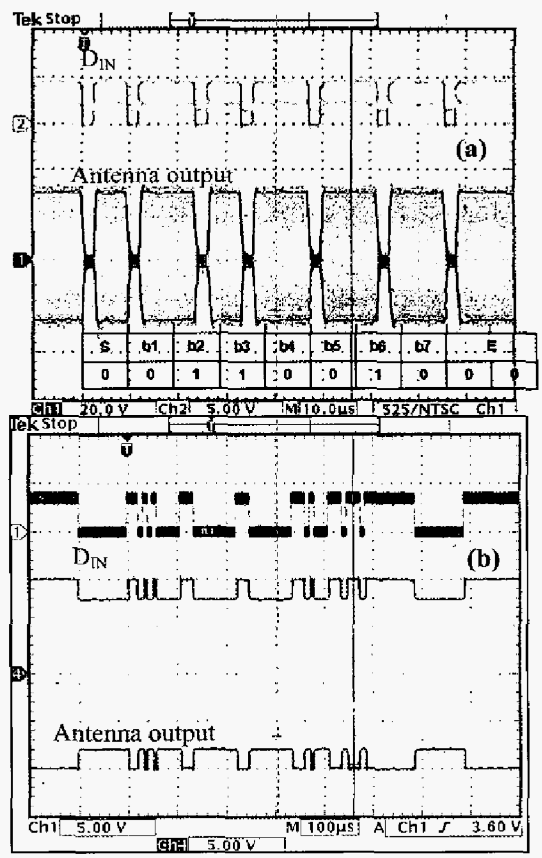

Fig.8 Measured waveforms of D_{IN} and output voltage on the antenna: (a) for 100%, and (b) for 10% ASK mode.

A test board of RFID interrogator was manufactured. Fig. 8 shows measured waveform of DIN and output on the loop antenna. The data of DIN are the frame data of the request-A command (REQA) encoded by modified Miller for ISO 14443 type-A mode and request-B (REQB) for 14443 type-B mode, respectively [1]. The measure waveforms show that the RF outputs on the antenna meet the ISO specification for rising and falling time, overshoot and undershoot, and AM modulation index under a supply voltage of 3.3 volt.

Fig 9 shows a subcarrier modulated signal on the antenna and corresponding R_{X_OUT} signal output of the receiving circuit. The ripple voltage on the antenna with subcarrier of 847 KHz is a response of RFID to a request command. The R_X signal was demodulated, so that each of ripple voltage was converted into corresponding pulse signal. The proposed analog front end circuit was operated in a 3.3 volt single supply voltage. Since the proposed circuit is design using standard CMOS with single supply voltage, it can be integrated with a digital part for a single chip design of RFID interrogator using a standard digital CMOS process.

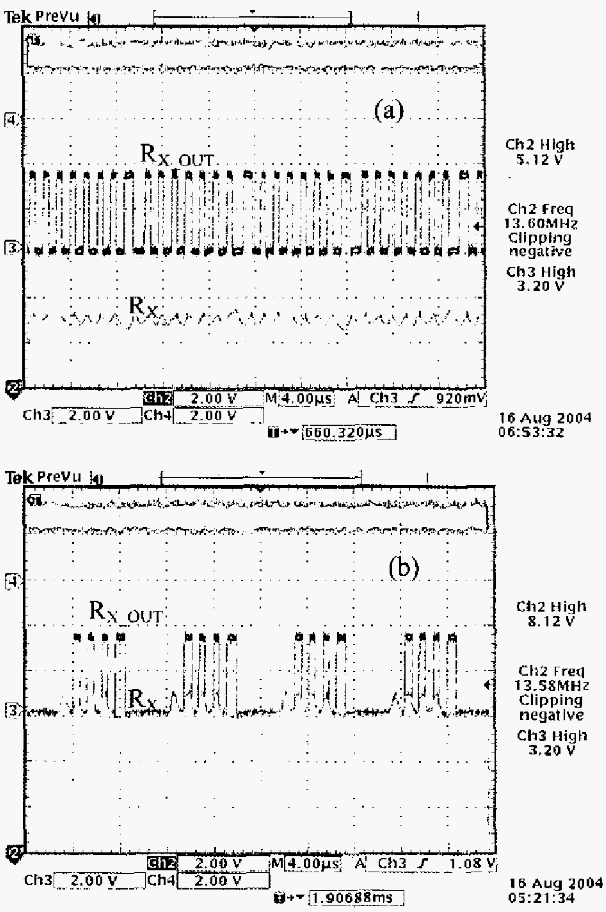

Fig. 9. Measured waveforms of load modulated antenna voltage (R_X) and demodulated R_{X_OUT} voltage for BPSK (a) and Manchester (b) subcarrier modulated input signal, respectively.

V. Summary and Conclusions

An analog front end circuit of 13.56MHz RFID interrogator compatible with ISO14443 type A/type B, ISO15693 and ISO18000-3 Mode 1 has been designed in full custom and fabricated by using a 0.35µm double poly CMOS process. The measured results demonstrated that the circuit has met the standard specifications. The results of this work can be provided as a reusable analog IPs for SoC solution of ISO compatible contactless card readers for integrating both analog and digital part into a single chip. The proposed circuit can be easily migrated to other CMOS processes for the IP based design, because the proposed circuit was designed and silicon proved using a 0.35µm CMOS process with single supply voltage of 3.3 volt which is available for the most of foundry companies

Acknowledgements

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